

Received 12 August 2025, accepted 26 September 2025, date of publication 7 October 2025, date of current version 6 January 2026.

Digital Object Identifier 10.1109/ACCESS.2025.3618519

## RESEARCH ARTICLE

# Resource-Aware Ensemble Framework for DDoS Detection in SDN via Novel Game-Theoretic Feature Selection

MOHAMED MEHDI TRIGUI<sup>1</sup>, MD SAKIBUL ISLAM<sup>2</sup>, MUHAMMAD DIKKO GAMBO<sup>2</sup>,  
MIR SUHAIL ALAM<sup>1</sup>, AND TAREK HELMY<sup>1,3</sup>

<sup>1</sup>Information and Computer Science Department, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

<sup>2</sup>Computer Engineering Department, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

<sup>3</sup>Interdisciplinary Research Center for Intelligent Secure Systems, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

Corresponding author: Md Sakibul Islam (g202321310@kfupm.edu.sa)

This work was supported by the King Fahd University of Petroleum and Minerals.

**ABSTRACT** Distributed Denial-of-Service (DDoS) attacks continue to threaten the reliability and performance of modern network infrastructures, particularly in Software-Defined Networking (SDN) environments, where the centralized controller represents a critical single point of failure. The growing complexity of DDoS vectors, including low-rate and zero-day attacks, calls for detection frameworks that are not only highly accurate but also generalizable and computationally efficient. This paper presents an enhanced ensemble-based machine learning framework designed for robust DDoS detection in SDN settings. At the core of our approach lies a novel hybrid feature selection method that combines Random Forest (RF) feature importance with the Banzhaf Power Index (BPI) from cooperative game theory. This RF-BPI method evaluates both the statistical relevance and marginal contribution of features, resulting in a reduced feature space that enhances detection performance while minimizing redundancy and computational cost. We systematically benchmark a range of classifiers, including XGBoost, Random Forest, Gradient Boosting, Support Vector Machines (SVM), k-Nearest Neighbors (KNN), AdaBoost, and Naïve Bayes, and integrate them into four voting-based ensemble configurations. Each ensemble model is independently trained and evaluated on two datasets: the CIC-DDoS2023 dataset, which reflects diverse and recent DDoS attack behaviors, and a flow-based SDN-specific dataset that emulates controller-targeted attack scenarios. All ensemble models achieve perfect classification performance (100% accuracy, precision, recall, and F1-score) on both datasets. Beyond accuracy, our evaluation also captures training time, inference latency, memory usage, and model size to assess the feasibility of real-time deployment. The results show that while some ensemble models complete training in under 5 seconds with memory footprints below 1 MB, others incur significantly higher resource costs, highlighting trade-offs between architectural complexity and runtime efficiency. This work demonstrates that a carefully designed ensemble IDS, strengthened by cooperative game-theoretic feature selection, can deliver high detection accuracy while maintaining flexibility and resource-awareness, making it suitable for deployment in production-grade SDN environments.

**INDEX TERMS** DDoS attack detection, software-defined networking, ensemble learning, feature selection, random forest, Banzhaf power index, voting classifier.

## I. INTRODUCTION

Distributed Denial-of-Service (DDoS) attacks continue to pose serious risks to the stability and security of digital

The associate editor coordinating the review of this manuscript and approving it for publication was Syed Islam<sup>1</sup>.

infrastructure. These attacks aim to exhaust network or computational resources by overwhelming services with malicious traffic, often disrupting availability and preventing access for legitimate users. Recent trends indicate a substantial increase in the frequency, scale, and sophistication of such attacks, with incidents now involving multi-terabit floods

and low-rate, stealthy traffic patterns that evade conventional detection systems [1], [2]. In particular, DDoS attacks are increasingly recognized as one of the most prevalent and destructive cyber threats faced by network operators worldwide [3].

Software-Defined Networking (SDN) has emerged as a powerful architectural solution that separates the control plane from the data plane, enabling centralized network management and programmability. While SDN offers clear advantages in terms of flexibility and traffic visibility, it also introduces new security risks. The centralized SDN controller becomes a highly attractive target for attackers, especially in DDoS scenarios. Adversaries can exploit vulnerabilities in control protocols or authentication mechanisms to flood the controller with malicious requests, effectively destabilizing the entire network [3], [4]. Given this centralized exposure, protecting the SDN control plane from advanced DDoS threats remains a critical research challenge [5].

In this context, it is essential to underline the dual role of SDN in DDoS defense. On the one hand, SDN provides fine grained flow visibility, centralized programmability, and a wide network coordination, all of which can significantly strengthen detection and mitigation capabilities. For example, flow level statistics enable early anomaly detection, and dynamic rule installation allows rapid throttling or redirection of malicious traffic. On the other hand, the very centralization that empowers SDN also introduces risks: the controller becomes a high value target and potential bottleneck under saturation. Therefore, any detection framework must not only be accurate but also lightweight enough to run next to the controller with minimal CPU, memory, and latency overhead, ensuring that security mechanisms do not themselves degrade controller performance [6]. Similar concerns have also been raised in broader machine learning research, where studies show that different classifiers vary substantially in their computational and energy requirements, highlighting sustainability as a critical evaluation dimension [7].

Traditional intrusion detection systems (IDSs) and standalone machine learning (ML) models have been applied to the DDoS problem, often achieving promising results in closed settings. However, these models frequently lack robustness and adaptability when applied to new attack variants or real-world traffic. Many ML-based detection systems perform well on the datasets they are trained on but show degraded performance when tested on unseen data or different environments [8], [9]. This overfitting problem, along with the rapid evolution of attack techniques, motivates the need for more resilient and adaptive detection frameworks that can generalize across diverse DDoS behaviors.

To address these challenges, this paper proposes an enhanced ensemble-based machine learning framework for DDoS detection in SDN environments. Our core contribution is a novel hybrid feature selection technique that combines Random Forest (RF) feature importance with the

Banzhaf Power Index (BPI) from cooperative game theory. While RF captures statistical relevance, BPI evaluates the marginal contribution of each feature to prediction outcomes. This dual strategy ensures that only the most impactful features are retained, thereby improving detection accuracy and reducing computational overhead [10]. Compared to traditional filter-based or wrapper-based selection methods, our RF-BPI approach offers both interpretability and model simplification.

We evaluate our method using two datasets: the CIC-DDoS2023 dataset, which contains modern, multi-vector DDoS attacks, and a flow-based SDN-specific dataset that reflects real controller-targeted behaviors. Separate models are trained and tested on each dataset to validate robustness. Four ensemble configurations are constructed using combinations of top-performing classifiers, including XGBoost, Random Forest, Gradient Boosting, Support Vector Machine (SVM), k-Nearest Neighbors (KNN), AdaBoost, and Naïve Bayes. All models achieve 100% accuracy, precision, recall, and F1-score.

Beyond predictive accuracy, our framework emphasizes the feasibility of deployment. We conduct detailed resource profiling of each ensemble model, measuring training time, inference latency, peak memory usage, and model size. These metrics are critical in SDN contexts, where real-time decisions must be made under strict resource constraints. For instance, we observe that while some ensemble configurations complete training in under five seconds and require less than 1 MB of memory, others achieve similar accuracy with significantly higher computational cost. Such insights enable informed trade-offs between accuracy and efficiency. Lastly, our study explores how the framework performs as the complexity of the classification task increases, such as in fine-grained multi-class DDoS detection. We observe that although accuracy remains high, classifier confusion tends to rise with the number of classes, indicating that deeper models or attention-based architectures may be needed in future work.

Building on these observations, the framework has the potential to be deployed as a controller-adjacent intrusion detection sidecar in SDN environments. In such a setup, it would continuously inspect flow statistics, apply lightweight feature selection, and employ an ensemble classifier to trigger mitigation actions such as throttling or blackholing suspicious flows. Operating alongside the controller rather than within it would help preserve controller stability while still enabling rapid and automated defense against evolving DDoS threats.

Overall, this paper delivers a practical and accurate DDoS detection solution for SDN by combining ensemble learning with game-theoretic feature engineering and full-spectrum resource evaluation. The findings contribute not only to advancing detection robustness but also to understanding how to optimize model selection for real-world SDN deployment.

## II. LITERATURE

Initial strategies for detecting Distributed Denial-of-Service (DDoS) attacks, both in conventional and Software-Defined Networking (SDN) environments, often relied on statistical analysis and fixed threshold rules. These approaches typically monitor flow-level characteristics such as packet rates, destination entropy, or connection counts collected from OpenFlow-enabled switches to flag anomalies. For instance, [11] proposed one of the earliest SDN-specific techniques that incorporated a flow collector and data plane feature extractor alongside a Self-Organizing Map (SOM) to distinguish between legitimate and malicious traffic [1]. While such systems can detect large-scale volumetric attacks, they are often ill-suited to low-rate or stealthy variants, and their reliance on manual tuning limits adaptability across different deployment environments.

With the growing sophistication of DDoS vectors, machine learning (ML)-based methods have gained prominence. Supervised learning algorithms such as support vector machines (SVM), decision trees, multilayer perceptrons (MLPs), and ensemble-based models have been extensively studied for SDN-based DDoS detection [12], [13]. These models are trained on labeled datasets to learn attack signatures or behavior patterns. Among them, Random Forest (RF) classifiers have consistently demonstrated strong results due to their ability to handle non-linear relationships and high-dimensional feature spaces. In one study, RF achieved 97% accuracy in detecting DDoS traffic within SDN environments [14], [15]. Similarly, near-perfect detection rates were reported for SVM and RF models when applied to benchmark datasets [12]; however, these evaluations were often confined to the training domain, revealing limitations when models face unseen attack types.

Advanced ML approaches have explored evolutionary enhancements. For example, [16] proposed an SVM-based method optimized using evolutionary algorithms, which yielded 99.9% accuracy during testing. Deep learning (DL) has also been adopted, with [17] deploying convolutional neural networks (CNNs) for SDN traffic classification, achieving over 98.8% accuracy with a false alarm rate as low as 0.02%. However, many of these solutions were evaluated on a single dataset, often the same one used during training, which raises concerns about generalizability to evolving or zero-day threats [8].

Recent research has emphasized the importance of learning traffic representations directly from data using deep architectures such as autoencoders, recurrent neural networks (RNNs), and LSTMs [2]. For instance, the LSTM-FCNet approach [18] demonstrates how temporal features extracted from channel state information (CSI) can significantly enhance detection in jamming-based DoS scenarios. Similarly, diverse ensemble methods [19] were combined with explainability techniques to improve robustness and transparency. These efforts underscore the broader applicability of hybrid and ensemble techniques, particularly

when performance must be maintained under resource constraints.

Despite their high detection accuracy, deep learning models often suffer from high computational complexity, long training times, and limited interpretability. In SDN contexts, where real-time decisions must be made at the controller level, large-scale DL architectures may not be feasible. To address this, some researchers have moved toward edge-deployable, lightweight models or have utilized programmable network hardware, such as P4-based data planes. For example, [20] presented an ML-enhanced P4 switch capable of handling part of the DDoS detection logic within the data plane itself, reducing controller load. While promising, these solutions require specialized infrastructure and often lack flexibility.

To overcome the limitations of single-model learners, ensemble and hybrid approaches have become increasingly popular. These methods integrate multiple base classifiers, typically employing bagging, boosting, or voting schemes to enhance detection reliability and mitigate overfitting [21]. For example, an ensemble-based SDN IDS that utilized the Data Plane Development Kit (DPDK) to manage high-throughput traffic demonstrated improved responsiveness by offloading detection to the data plane. Similarly, [5] demonstrated an ensemble of bagged RF and boosted decision trees that achieved near-perfect detection across multiple datasets. Another approach by [2] combined two-stage feature selection with ensemble learning to reduce false positives while retaining high true positive rates. Collectively, these studies confirm that ensembles can provide robust and adaptive defenses against DDoS attacks, especially when paired with effective feature engineering.

Feature selection remains a crucial aspect of intrusion detection model design. Modern network datasets can include hundreds of features ranging from flow statistics to protocol metadata, many of which may be redundant or irrelevant. Reducing dimensionality not only speeds up training and inference but can also enhance accuracy. Prior works have utilized mutual information, entropy-based filters, recursive feature elimination (RFE), and evolutionary algorithms to select optimal subsets [22]. In one such study, feature reduction using statistical heuristics was shown to improve both precision and runtime when training XGBoost classifiers on DDoS datasets.

Despite this progress, several key limitations persist. Many past studies have relied on outdated datasets such as CIC-DDoS2019 or synthetic traffic, which may not reflect the complexity of modern attack strategies. Furthermore, most works focus exclusively on detection accuracy, with little to no attention paid to real-world performance metrics such as memory usage, inference latency, or model size. This gap makes deployment in operational SDN controllers risky or inefficient.

To address these gaps, our work builds on recent advances in ensemble learning and game-theoretic feature

selection. We utilize the CIC-DDoS2023 dataset, which offers current and diverse attack scenarios, and combine it with an SDN flow dataset to demonstrate cross-environment robustness. In addition, our framework explicitly profiles each ensemble's resource footprint, including training time, memory consumption, and storage size, ensuring practical deployability. These contributions set our approach apart by emphasizing not only detection robustness but also resource efficiency and real-world readiness.

### III. METHODOLOGY

#### A. DATASET DESCRIPTION

To evaluate the performance and generalizability of the proposed DDoS detection framework, two datasets were used: a flow-level SDN dataset and the CIC-DDoS2023 dataset.

##### 1) SDN FLOW-BASED DATASET

The first dataset, titled "DDoS Attack Detection Classification," contains flow-level network traffic collected in a SDN environment. It includes 104,345 records and 23 features, such as packet count, byte count, flow duration, packet rates, and protocol types. Each record describes a communication flow between a source and destination node, specified by IP addresses and port numbers, along with associated traffic statistics.

This dataset includes labels indicating whether each flow is normal or represents an attack, making it suitable for supervised learning tasks. The variety of traffic features allows for detailed analysis of SDN traffic patterns and provides a useful base for building and evaluating intrusion detection models.

##### 2) CIC-DDoS2023 DATASET

The second dataset is CIC-DDoS2023, a recent benchmark designed to reflect a wide range of modern DDoS attack behaviors. We took a subset of the main dataset that contains 1,754,109 rows and 47 features, covering various aspects of network traffic, including flow duration, protocol type, transmission rate, packet inter-arrival time (IAT), and TCP flag statistics. It also includes statistical features such as standard deviation, variance, and covariance.

Each row corresponds to a network flow and is labeled with one of several traffic classes, such as DDoS, Mirai, or Spoofing. Sample features include *flow\_duration*, *protocol\_type*, *rate*, *drate*, *weight*, and *magnitude*. The dataset is well-suited for binary and multi-class classification tasks and supports experiments in both attack detection and feature selection research.

#### B. CLASSIFICATION ARCHITECTURES

The proposed detection framework leverages a range of well-established supervised machine learning classifiers, each chosen for its ability to handle high-dimensional,

network-based traffic data. These models are individually trained and evaluated to benchmark their classification performance before being integrated into ensemble configurations.

##### 1) XGBoost CLASSIFIER

XGBoost is a scalable and efficient gradient boosting framework based on decision trees. It is particularly well-suited for learning complex, non-linear patterns in tabular data. Due to its regularization mechanisms and parallelized training, XGBoost tends to outperform traditional gradient boosting models in both speed and accuracy. In this work, it serves as a key component in both individual and ensemble detection pipelines [23].

##### 2) SUPPORT VECTOR MACHINE (SVM)

SVMs are known for their ability to find optimal decision boundaries by maximizing the margin between classes. They perform well in high-dimensional feature spaces and are especially effective when the number of samples is small relative to the number of features. Kernel functions allow SVMs to capture non-linear relationships, which is useful in classifying complex network behaviors [16].

##### 3) RANDOM FOREST CLASSIFIER

Random Forests are ensemble models that aggregate predictions from multiple decision trees to improve classification stability and reduce overfitting. By randomly selecting subsets of features and data points during training, they introduce useful variance and are particularly effective in high-dimensional settings such as network traffic classification [24].

##### 4) GRADIENT BOOSTING CLASSIFIER

Gradient Boosting sequentially constructs decision trees to correct the errors of prior models. Unlike bagging techniques such as Random Forest, gradient boosting focuses on learning from the mistakes of the previous model, leading to more refined decision boundaries. In this framework, it complements other learners by improving classification precision, particularly in imbalanced datasets [25].

##### 5) ADABOOST CLASSIFIER

AdaBoost works by assigning greater weight to misclassified samples in each iteration, allowing subsequent learners to focus more on difficult cases. Its performance in imbalanced and noisy data environments makes it a valuable addition to our classifier pool. As a boosting-based algorithm, it works well with weak learners and contributes to ensemble diversity [26].

##### 6) GAUSSIAN NAIVE BAYES

Naive Bayes classifiers rely on probabilistic reasoning and feature independence assumptions. The Gaussian variant assumes that the feature distribution is normal. While simple,

it offers low computational cost and fast prediction, making it an ideal lightweight baseline. In our experiments, it is included both as a standalone model and within ensemble configurations [27]

#### 7) k-NEAREST NEIGHBORS (K-NN)

The k-NN algorithm classifies samples based on the majority label among the k most similar instances in the training set, as measured by distance metrics such as Euclidean distance. Though non-parametric and computationally intensive at inference time, it provides a useful contrast to tree-based learners and is especially interpretable in small feature spaces.

Each of these classifiers was initially trained and tested independently using both the SDN and CIC-DDoS2023 datasets. The highest-performing models were then selected for ensemble construction to exploit complementary strengths and improve overall robustness.

### C. BANZHAF POWER INDEX FOR FEATURE SELECTION

The Banzhaf Power Index is a cooperative game-theoretic measure that quantifies the relative importance or influence of a player, in this case, a feature, within all possible coalitions (feature subsets). Originally developed in the context of voting theory to evaluate political power, BPI has been adapted for feature selection in machine learning to assess the marginal impact of each feature on model performance.

Let  $F = \{f_1, f_2, \dots, f_n\}$  denote the set of all features and  $M$  be a predictive model. The BPI of a feature  $f_i$  is based on its marginal contribution to classification performance across all subsets of features.

- 1) Marginal Contribution: For any subset  $S \subseteq F \setminus \{f_i\}$ , the marginal contribution of  $f_i$  is given by:

$$\Delta_{f_i}(S) = \text{Perf}(S \cup \{f_i\}) - \text{Perf}(S),$$

where  $\text{Perf}(S)$  is a chosen performance metric (e.g., accuracy) obtained by training the model on a feature subset  $S$ .

- 2) Banzhaf Value: The Banzhaf value  $\beta_{f_i}$  is the average marginal contribution over all subsets:

$$\beta_{f_i} = \frac{1}{2^{n-1}} \sum_{S \subseteq F \setminus \{f_i\}} \Delta_{f_i}(S).$$

This formula assumes equal probability of every subset being chosen.

- 3) Normalized Banzhaf Index: To facilitate comparison across features, the BPI is normalized:

$$\text{BPI}(f_i) = \frac{\beta_{f_i}}{\sum_{j=1}^n \beta_{f_j}},$$

ensuring that  $\sum_{i=1}^n \text{BPI}(f_i) = 1$ .

The BPI provides a global view of feature importance by estimating the average influence of each feature across all possible combinations. Features with high BPI values are those that frequently alter model predictions when included

or excluded, suggesting they carry critical decision-making power. Conversely, features with near-zero BPI values are likely redundant or irrelevant and can be safely removed to improve efficiency without significant loss in performance.

In this study, BPI was used to refine feature selection for DDoS detection. Due to the large size of the test set, a balanced stratified sample was first constructed from the original  $X_{\text{test}}$ . Samples were randomly drawn from each class (normal and attack) to form a smaller, representative test subset. To estimate BPI scores, a trained classification model was used to predict labels for each sample. The marginal impact of each feature was assessed by flipping its value in a given row and observing whether this flip changed the model's prediction. The BPI score for a feature was defined as the proportion of samples where such a change in prediction occurred, averaged over all rows.

Formally, let  $x \in X_{\text{sample}}$  be a row and  $f_i$  a feature. For each sample:

$$\text{BPI}_i = \frac{1}{|X|} \sum_{x \in X} \mathbb{I}[M(x) \neq M(x^{(i)})],$$

where  $x^{(i)}$  is the same input with the  $i$ -th feature perturbed (flipped), and  $\mathbb{I}[\cdot]$  is the indicator function.

The resulting BPI scores were normalized and ranked. Features with higher BPI values were retained for training, while those with negligible impact were excluded. This process enabled both improved accuracy and reduced computational overhead, especially when feeding the selected features into ensemble classifiers.

### D. ATTACK CLASSIFICATION USING INDIVIDUAL ML MODELS

Before constructing ensemble models, we evaluated each classifier to establish a performance baseline. This step is crucial for identifying models that demonstrate high accuracy and are thus suitable for inclusion in ensemble configurations. The experiments were conducted using the feature subsets selected by the Banzhaf Power Index (BPI), ensuring uniformity across all comparisons. Figures 1 and 2 present the classification accuracy of eleven machine learning classifiers evaluated on the two datasets: SDN and CIC-DDoS2023.

As shown in the figures, tree-based models such as Random Forest, XGBoost, Gradient Boosting, and Extra Trees consistently outperformed other classifiers across both datasets. On the CIC-DDoS2023 dataset, most models achieved nearly perfect accuracy, indicating that the attack patterns in this dataset are relatively well-defined and separable. XGBoost recorded the highest accuracy, followed closely by Random Forest and Gradient Boosting.

In contrast, the SDN dataset exhibited greater variance in performance. Traditional models such as Logistic Regression, Naive Bayes, and Neural Networks showed significantly lower accuracy, suggesting that this dataset presents more subtle or overlapping class distributions.

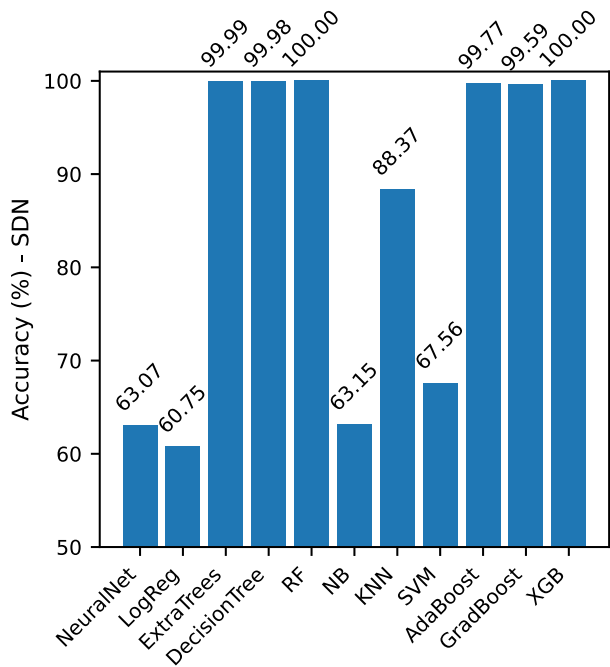


FIGURE 1. Classification accuracy of individual models on the SDN dataset.

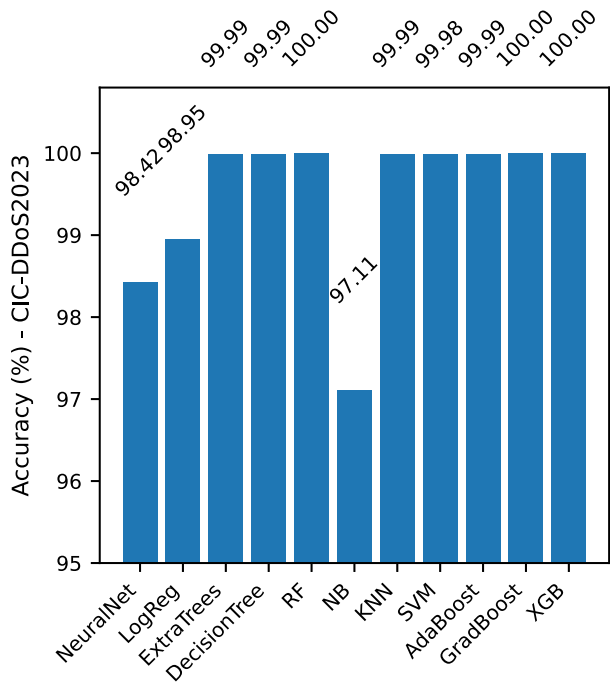


FIGURE 2. Classification accuracy of individual models on the CIC-DDoS2023 dataset.

SVM and k-NN demonstrated moderate performance but were still outperformed by tree-based learners.

These differences emphasize the need for ensemble approaches that combine the strengths of multiple models.

TABLE 1. Base classifiers in each ensemble configuration.

| Ensemble Model | Included Classifiers                                                  |
|----------------|-----------------------------------------------------------------------|
| clf1           | XGBoost, Random Forest, Gradient Boosting                             |
| clf2           | XGBoost, Random Forest, AdaBoost, k-NN                                |
| clf3           | XGBoost, Random Forest, Gradient Boosting, SVM                        |
| clf4 (Full)    | XGBoost, Random Forest, Gradient Boosting, AdaBoost, SVM, Naive Bayes |

By integrating the highest-performing base learners, it becomes possible to achieve both robust detection and improved generalization across diverse traffic scenarios. The ensemble configurations described in the next section were designed with these observations in mind.

E. ENSEMBLE CLASSIFICATION MODELS

While several individual classifiers achieved high performance, differences in accuracy across datasets, especially the SDN dataset, highlighted the potential for model-specific biases or sensitivity to feature distributions. To enhance robustness and generalization, we implemented four ensemble models that aggregate predictions from multiple high-performing base classifiers using a majority voting strategy.

The intuition behind using ensemble learning is that individual classifiers may make different types of errors; by combining their decisions, the ensemble can mitigate weaknesses and reinforce consistent signals. This is especially useful in cybersecurity applications, where the variability of traffic patterns and attack types makes overfitting or dataset optimization a common issue.

All ensembles use a hard voting mechanism, where each base learner casts a vote for the predicted class, and the final prediction is determined by majority rule. This approach avoids the need to calibrate probability outputs and is computationally simple during inference. The composition of each ensemble is shown in Table 1.

Each ensemble was trained independently on both the SDN and CIC-DDoS2023 datasets, using RF-BPI-selected features and a preprocessing pipeline, as individual classifiers. The chosen base models were selected based on their high accuracy in the earlier stage and their complementary decision characteristics. Notably, clf4 integrates both complex models (e.g., XGBoost, SVM) and simpler learners (e.g., Naïve Bayes) to maximize diversity. In the next section, we evaluate the detection performance of each ensemble as well as their training time, inference latency, and memory footprint to assess their practical applicability in SDN environments.

F. RESOURCE PROFILING METHODOLOGY

In addition to classification performance, evaluating the resource efficiency of a model is critical for real-world deployment, particularly in resource-constrained environments such as Software-Defined Networking (SDN) controllers. An intrusion detection model deployed at the controller level must not introduce significant latency or

memory overhead that could interfere with core network operations.

To assess deployment feasibility, we measured the computational footprint of each ensemble configuration using four key metrics:

- **Training Time:** The wall-clock duration required to train the model on the training set. This was measured using Python's `time.perf_counter()` before and after training.
- **Inference Time:** The time required for the trained model to classify the entire test set, captured using the same high-resolution timing method.
- **Memory Usage:** Peak memory consumption was recorded during both training and inference phases using the `memory_usage()` function from the `memory_profiler` package.
- **Model Size:** After training, each ensemble was serialized using the `joblib` library. The resulting file size in megabytes (MB) was used to quantify storage requirements.

These measurements help identify the trade-offs between model complexity and operational efficiency. For example, while some ensemble configurations may achieve high accuracy, their large memory footprint or slow inference speed may make them unsuitable for time-sensitive SDN deployments.

All metrics were collected under consistent hardware and software settings to ensure fair comparison across ensemble models. The detailed resource usage values and their impact on model selection are presented in the results section.

#### G. PERFORMANCE EVALUATION KEY INDICATORS

To assess the effectiveness and reliability of the proposed models, standard performance metrics derived from the confusion matrix were used. These metrics provide a detailed view of classification quality and are especially relevant for evaluating models on security-related tasks such as intrusion detection.

The confusion matrix summarizes prediction outcomes in terms of four fundamental states:

- **True Positive (TP):** Malicious traffic correctly identified as an attack.
- **True Negative (TN):** Benign traffic correctly identified as normal.
- **False Positive (FP):** Benign traffic incorrectly classified as an attack.
- **False Negative (FN):** Malicious traffic incorrectly classified as benign.

These four outcomes form the basis for several widely used performance metrics:

- **Accuracy:** The proportion of total correct predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** The proportion of correct positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall (Sensitivity):** The proportion of actual positive instances that are correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-Score:** The harmonic mean of precision and recall. It balances the trade-off between false positives and false negatives, making it especially useful for imbalanced datasets.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Together, these metrics provide a well-rounded evaluation of model performance. In this study, particular emphasis is placed on precision and F1-score, as they offer insight into both the model's accuracy and its ability to minimize false positives, a crucial consideration in intrusion detection contexts.

## IV. RESULTS AND EVALUATION

This section presents the evaluation outcomes of the proposed ensemble classifiers across two primary axes: (1) classification and system resource performance on SDN and CIC-DDoS2023 datasets, and (2) comparative analysis with state-of-the-art DDoS detection methods from the literature.

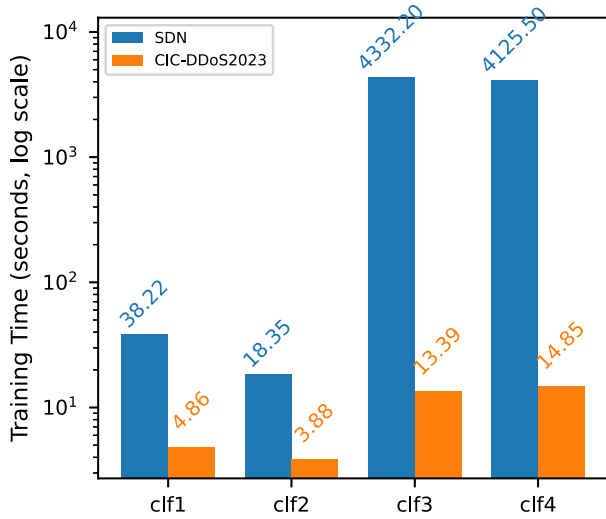
### A. CLASSIFICATION AND RESOURCE PERFORMANCE

Each ensemble model (clf1 to clf4) was evaluated using the standard performance metrics described in Section III-G. Across both the SDN and CIC-DDoS2023 datasets, all configurations achieved 100% accuracy, precision, recall, and F1-score. These consistent results indicate the stability and effectiveness of the voting-based ensemble approach when applied to heterogeneous DDoS detection scenarios.

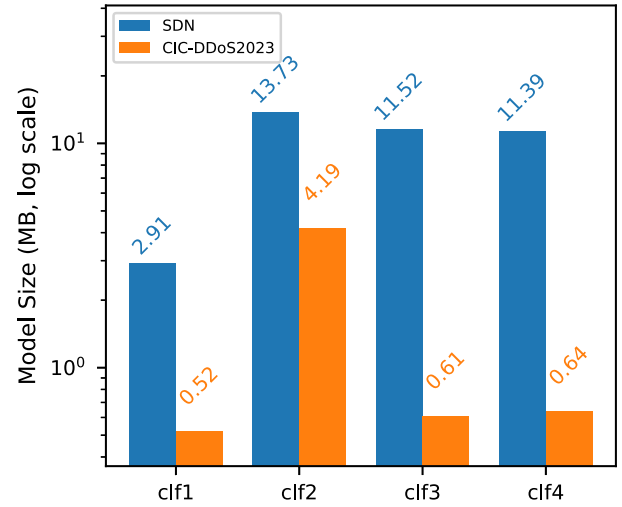
Although classification accuracy was uniform across models, significant differences were observed in terms of computational resource usage. This reinforces the importance of evaluating not just detection performance but also system-level efficiency, especially when the intended deployment environment involves real-time constraints, such as SDN controllers or edge nodes. Figures 3 to 5 present a detailed comparison of each ensemble model across three key resource metrics: training time, inference latency, and model size.

As shown in Figure 3, clf1 and clf2 trained significantly faster than clf3 and clf4, particularly on the SDN dataset. Clf3 and clf4 incurred over 4000 seconds of training time due to the inclusion of SVM, which is known to be computationally expensive on large datasets. In contrast, on CIC-DDoS2023, all models trained quickly (under 15 seconds), reflecting the dataset's smaller size and more consistent structure.

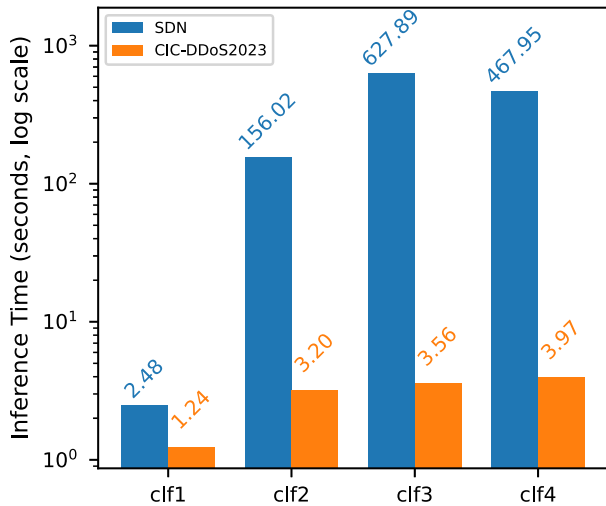
Figure 4 highlights a substantial variation in prediction latency. Clf2, which includes k-Nearest Neighbors (k-NN), required over 150 seconds to classify the SDN test set. This is a known limitation of instance-based learning, where each prediction involves distance calculations against all training points. Clf3 and clf4 also demonstrated increased inference



**FIGURE 3.** Training time comparison across ensemble models and datasets.



**FIGURE 5.** Model size comparison across ensemble models and datasets.



**FIGURE 4.** Inference time comparison across ensemble models and datasets.

time due to the inclusion of SVM, while clf1 maintained low latency across both datasets.

Model size comparisons in Figure 5 further reinforce the differences between ensemble configurations. While clf1 and clf3 remained lightweight (under 1 MB) on CIC-DDoS2023, their counterparts trained on SDN grew significantly larger due to increased complexity and dataset size. Clf2's model size was especially high, again reflecting the storage burden of k-NN structures.

#### Summary of Findings:

- Clf1 offers the best balance between performance and efficiency and is well-suited for real-time or embedded deployments.

- Clf2 is efficient during training but suffers from slow inference due to k-NN.
- Clf3 and Clf4, while accurate, are resource-heavy and best suited for offline detection or analysis tasks.

These observations underscore the importance of resource-aware model selection, especially in SDN environments where processing and memory capacity are constrained.

#### B. COMPARATIVE ANALYSIS WITH EXISTING WORKS

To contextualize the performance of the proposed ensemble-based detection framework, we compared it with several recent DDoS detection approaches reported in the literature. Table 2 provides a summary of techniques used, datasets evaluated, and key performance metrics (accuracy and F1-score).

A number of prior works have reported strong results on classic datasets such as KDDCup99, DARPA, CICIDS2017, and CIC-DDoS2019. For instance, Alduailij et al. utilized a combination of ensemble classifiers (RF, KNN, GB, LR) along with a voting mechanism, achieving 99% accuracy on CICIDS2017 and CIC-DDoS2019. Similarly, Becerra-Suarez et al. (2 combined tree-based learners and deep networks to achieve 99.97% accuracy and 99.98% F1-score. Other studies have explored deep learning methods such as CNNs and BiLSTMs [28], or autoencoder-based architectures [29] for improved detection.

While these results are promising, most studies are limited to older datasets and do not evaluate performance on more recent and diverse attack scenarios. Many also lack resource profiling, making it difficult to assess practical deployability. For example, although deep learning models like BiLSTM or autoencoder-based networks can achieve high accuracy, they often require substantial training time, GPU support, and memory resources—constraints that make them less suitable for real-time SDN controller environments.

**TABLE 2.** Performance comparison with other works.

| Study                        | Detection Method                    | Dataset Used             | Accuracy       | F1-Score       |
|------------------------------|-------------------------------------|--------------------------|----------------|----------------|
| Alduailij et al. (2022)      | RF, KNN, GB, LR, WVE                | CICIDS2017, CIC-DDoS2019 | 99%            | –              |
| Saif et al. (2022)           | RF, KNN, SVM, NB, J48               | WSN-DS                   | 99.67%         | –              |
| Alghazzawi et al. (2021)     | CNN + BiLSTM                        | CIC-DDoS2019             | 94.52%         | –              |
| Abubakar et al. (2020)       | SVM + SNORT IPS                     | KDDCup99, DARPA          | 97.0%          | –              |
| Becerra-Suarez et al. (2024) | RF, DT, ADA, XGB, DNN               | CIC-DDoS2019             | 99.97%         | 99.98%         |
| Xu and Fan (2022)            | Autoencoder + XGBoost               | UNSW-NB15, CICIDS2017    | 95.11–99.92%   | –              |
| <b>Ours (clf1–clf4)</b>      | Voting Ensemble (XGB, RF, GB, etc.) | CIC-DDoS2023, SDN        | <b>100.00%</b> | <b>100.00%</b> |

In contrast, our proposed framework offers several key advantages:

- **Use of Recent and Diverse Dataset:** Unlike prior works, our evaluation includes CIC-DDoS2023—a large-scale, realistic dataset that captures modern DDoS attack vectors with more granular flow-level statistics. This ensures that our model is tested against more complex, multi-stage, and stealthy attacks that better reflect current threat landscapes.
- **Cross-Dataset Generalization:** The ensemble models were validated on both CIC-DDoS2023 and an SDN-specific dataset. Achieving 100% accuracy and F1-score on both datasets underscores the generalizability of the method and its resistance to dataset-specific high performance, an issue not addressed in many prior studies.
- **Lightweight and Modular Design:** While maintaining top-tier accuracy, the proposed ensemble clf1 exhibits low model size, short training time, and fast inference. This makes the solution viable for deployment on SDN controllers or at the edge, unlike heavyweight deep learning models.
- **Comprehensive Evaluation:** In addition to predictive performance, we analyzed training time, inference time, memory usage, and model size. These are critical dimensions for any IDS meant to run in real-time, high-availability networking systems.

It is worth noting that while several previous studies reported close to 99–100% accuracy, none achieved a perfect classification score across multiple datasets, nor did they quantify the system cost of their models in terms of memory and latency [30], [31]. By combining a novel game-theoretic feature selection mechanism with a multi-model voting strategy and a resource-aware evaluation process, our framework pushes the boundary of both detection accuracy and operational practicality.

These comparisons reinforce that our work contributes not only in terms of performance, but also in terms of real-world readiness, a critical requirement in modern SDN deployments.

## C. DISCUSSION

The evaluation results demonstrate that all ensemble models (clf1 to clf4) achieved perfect classification performance

across both the CIC-DDoS2023 and SDN datasets. This confirms the effectiveness of ensemble learning for DDoS detection, particularly when models are selected based on their complementary strengths.

However, the comparison of resource metrics reveals important practical differences. While clf3 and clf4 achieved the same level of accuracy as the simpler configurations, they required significantly higher computational resources. In contrast, clf1 and clf2 offered an ideal trade-off between performance and efficiency. Clf1, in particular, consistently showed fast training and low inference times with small model size, making it suitable for real-time deployment in SDN environments with limited computational capacity.

These findings suggest that, in practice, lightweight ensemble configurations such as clf1 should be preferred when operating under resource constraints. More complex ensembles may still be valuable in offline analysis scenarios or environments where detection latency is less critical.

Furthermore, the superior generalization performance across two different datasets, CIC-DDoS2023 and SDN dataset, demonstrates the robustness of the proposed framework. Unlike prior works that report results on a single benchmark, this study validates the model across multiple real-world conditions, addressing a key limitation in earlier research.

Future work will explore broader testing and deployment. We plan to evaluate additional ensemble combinations under diverse traffic patterns to identify setups that best balance detection quality and resource usage. The framework will also be tested in varied deployment scenarios, including sidecar operation near the controller, placement at edge nodes, and partial offloading to programmable switches, to examine its impact on latency and scalability. Another step is to extend beyond offline datasets by using large-scale testbeds or emulated SDN environments for results that more closely mirror real-world conditions. Finally, the use of production traffic traces will allow us to assess the practicality of the framework in live SDN systems.

## V. CONCLUSION

This paper introduced a robust and resource-aware ensemble-based intrusion detection framework for detecting DDoS

attacks in Software-Defined Networking environments. We proposed a novel feature selection approach that combines Random Forest feature importance with the Banzhaf Power Index (BPI) to identify and retain the most influential features.

Four ensemble configurations were evaluated, each constructed using top-performing classifiers such as XGBoost, Random Forest, Gradient Boosting, AdaBoost, and SVM. The models were assessed on CIC-DDoS2023 and the SDN traffic dataset using standard performance metrics and a detailed resource profiling methodology.

All ensembles achieved perfect classification accuracy. However, significant variation was observed in training time, inference time, memory usage, and model size. Among the configurations, clfl emerged as the most efficient and deployment-ready, combining high accuracy with minimal resource overhead.

This work addresses several limitations of existing IDS approaches by providing both a generalizable detection strategy and a realistic assessment of computational cost. Future work may focus on extending the framework to multi-class DDoS attack detection, incorporating online learning techniques, and exploring deployment within distributed or federated SDN architectures.

## REFERENCES

- [1] A. J. Akroma and E. O. K. Mohamed, "Analysis of DDoS attacks and development of software solutions using machine learning for detection and mitigation," *Bani Waleed Univ. J. Humanities Appl. Sci.*, vol. 10, no. 2, pp. 39–50, Apr. 2025.
- [2] M. A. Hossain and M. S. Islam, "Enhancing DDoS attack detection with hybrid feature selection and ensemble-based classifier: A promising solution for robust cybersecurity," *Meas., Sensors*, vol. 32, Apr. 2024, Art. no. 101037.
- [3] M. S. Farooq, S. Riaz, and A. Alvi, "Security and privacy issues in software-defined networking (SDN): A systematic literature review," *Electronics*, vol. 12, no. 14, p. 3077, Jul. 2023.
- [4] M. P. Novaes, L. F. Carvalho, J. Lloret, and M. L. Proença, "Adversarial deep learning approach detection and defense against DDoS attacks in SDN environments," *Future Gener. Comput. Syst.*, vol. 125, pp. 156–167, Dec. 2021.
- [5] A. A. Alashhab, M. S. Zahid, B. Isyaku, A. A. Elnour, W. Nagmeldin, A. Abdelmaboud, T. A. A. Abdullah, and U. D. Maiwada, "Enhancing DDoS attack detection and mitigation in SDN using an ensemble online machine learning model," *IEEE Access*, vol. 12, pp. 51630–51649, 2024.
- [6] Y. Su, D. Xiong, K. Qian, and Y. Wang, "A comprehensive survey of distributed denial of service detection and mitigation technologies in software-defined network," *Electronics*, vol. 13, no. 4, p. 807, Feb. 2024.
- [7] M. S. Islam, S. N. Zisad, A.-L. Kor, and M. H. Hasan, "Sustainability of machine learning models: An energy consumption centric evaluation," in *Proc. Int. Conf. Electr., Comput. Commun. Eng. (ECCE)*, Feb. 2023, pp. 1–6.
- [8] A. Zaman, S. A. Khan, N. Mohammad, A. A. Ateya, S. Ahmad, and M. A. ElAffendi, "Distributed denial of service attack detection in software-defined networks using decision tree algorithms," *Future Internet*, vol. 17, no. 4, p. 136, Mar. 2025.
- [9] N. Prazeres, R. L. D. C. Costa, L. Santos, and C. Rabadão, "Engineering the application of machine learning in an IDS based on IoT traffic flow," *Intell. Syst. Appl.*, vol. 17, Feb. 2023, Art. no. 200189.
- [10] A. Mortazavi and M. H. Moattar, "Robust feature selection from microarray data based on cooperative game theory and qualitative mutual information," *Adv. Bioinf.*, vol. 2016, pp. 1–16, Mar. 2016.
- [11] R. Braga, E. Mota, and A. Passito, "Lightweight DDoS flooding attack detection using NOX/OpenFlow," in *Proc. IEEE Local Comput. Netw. Conf.*, Oct. 2010, pp. 408–415.
- [12] H. Z. Rui, T. Y. Chien, L. X. Ee, C. W. San, and L. T. Yi, "Comparison of the use of support vector machine (SVM) & random forest algorithms (RF) for DDOS attack detection," *Int. J. Res. Innov. Social Sci.*, vol. IX, no. I, pp. 1126–1138, 2025.
- [13] V. Thalapala, A. K. Reddy, and K. Guravaiah, "MLADAD: Machine learning algorithms analysis on DDoS attack detection in SDN-IoT networks," *Proc. Comput. Sci.*, vol. 260, pp. 1121–1128, Sep. 2025.
- [14] S. Rajesh, M. Clement, S. B. Sooraj, S. H. A. Shifan, and J. Johnson, "Real-time DDoS attack detection based on machine learning algorithms," in *Proc. Yukthi 2021–Int. Conf. Emerg. Trends Eng.—GEC Kozhikode*, Kerala, India, Sep. 2021. [Online]. Available: <https://ssrn.com/abstract=3974241>
- [15] I. Ortet Lopes, D. Zou, F. A. Ruambo, S. Akbar, and B. Yuan, "Towards effective detection of recent DDoS attacks: A deep learning approach," *Secur. Commun. Netw.*, vol. 2021, pp. 1–14, Nov. 2021.
- [16] K. S. Sahoo, B. K. Tripathy, K. Naik, S. Ramasubbareddy, B. Balusamy, M. Khari, and D. Burgos, "An evolutionary SVM model for DDOS attack detection in software defined networks," *IEEE Access*, vol. 8, pp. 132502–132513, 2020.
- [17] N. Ahuja, G. Singal, D. Mukhopadhyay, and N. Kumar, "Automated DDOS attack detection in software defined networking," *J. Netw. Comput. Appl.*, vol. 187, Aug. 2021, Art. no. 103108.
- [18] S. Mohan, S. Nandhini, and K. Gunaseelan, "Detection of denial-of-service attack using a novel hybrid learning technique," *J. Inf. Secur. Appl.*, vol. 92, Jul. 2025, Art. no. 104081.
- [19] A. Oad, S. S. Abbas, A. Zafar, B. A. Akram, F. Dong, M. S. H. Talpur, and M. Uddin, "Plant leaf disease detection using ensemble learning and explainable AI," *IEEE Access*, vol. 12, pp. 156038–156049, 2024.
- [20] F. Musumeci et al., "Machine-learning-enabled DDoS attacks detection in P4 programmable networks," *J. Netw. Syst. Manage.*, vol. 30, 2022, Art. no. 21, doi: [10.1007/s10922-021-09633-5](https://doi.org/10.1007/s10922-021-09633-5).
- [21] J. E. Varghese and B. Muniyal, "An efficient IDS framework for DDoS attacks in SDN environment," *IEEE Access*, vol. 9, pp. 69680–69699, 2021.
- [22] P. Araujo, A. Silva, N. Junior, F. Cabrini, A. Santos, A. Guelfi, and S. Kofuji, "Impact of feature selection methods on the classification of DDoS attacks using XGBoost," *J. Commun. Inf. Syst.*, vol. 36, no. 1, pp. 200–214, 2021.
- [23] Ismail, M. I. Mohmand, H. Hussain, A. A. Khan, U. Ullah, M. Zakarya, A. Ahmed, M. Raza, I. U. Rahman, and M. Haleem, "A machine learning-based classification and prediction technique for DDoS attacks," *IEEE Access*, vol. 10, pp. 21443–21454, 2022.
- [24] M. Alduailij, Q. W. Khan, M. Tahir, M. Sardaraz, M. Alduailij, and F. Malik, "Machine-learning-based DDoS attack detection using mutual information and random forest feature importance method," *Symmetry*, vol. 14, no. 6, p. 1095, May 2022.
- [25] H. A. Alamri and V. Thayanathan, "Bandwidth control mechanism and extreme gradient boosting algorithm for protecting software-defined networks against DDoS attacks," *IEEE Access*, vol. 8, pp. 194269–194288, 2020.
- [26] A. R. Javed, Z. Jalil, S. A. Moqurrah, S. Abbas, and X. Liu, "Ensemble AdaBoost classifier for accurate and fast detection of botnet attacks in connected vehicles," *Trans. Emerg. Telecommun. Technol.*, vol. 33, no. 10, Aug. 2020.
- [27] S. Naiem, A. E. Khedr, A. M. Idrees, and M. I. Marie, "Enhancing the efficiency of Gaussian Naïve Bayes machine learning classifier in the detection of DDOS in cloud computing," *IEEE Access*, vol. 11, pp. 124597–124608, 2023.
- [28] D. Alghazzawi, O. Bamasag, H. Ullah, and M. Z. Asghar, "Efficient detection of DDoS attacks using a hybrid deep learning model with improved feature selection," *Appl. Sci.*, vol. 11, no. 24, p. 11634, Dec. 2021.
- [29] W. Xu and Y. Fan, "Intrusion detection systems based on logarithmic autoencoder and XGBoost," *Secur. Commun. Netw.*, vol. 2022, Apr. 2022, Art. no. 9068724.
- [30] M. D. S. Islam, A. Yusuf, M. D. Gambo, and A. Y. Barnawi, "A novel few-shot ML approach for intrusion detection in IoT," *Arabian J. Sci. Eng.*, vol. 50, no. 10, pp. 7765–7779, Dec. 2024.
- [31] F. L. Becerra-Suarez, I. Fernández-Roman, and M. G. Forero, "Improvement of distributed denial of service attack detection through machine learning and data processing," *Mathematics*, vol. 12, no. 9, p. 1294, Apr. 2024.



interests include artificial intelligence, cybersecurity, machine learning, and DevOps automation.

**MOHAMED MEHDI TRIGUI** received the bachelor's degree in computer engineering from the National Engineering School of Sfax (ENIS), Tunisia. He is currently pursuing the master's degree in computer science with the King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia, where he has also submitted his thesis proposal. He has professional experience as a Software Developer, a DevOps Engineer, and a Graduate Research Assistant. His research



field. His research interests include artificial intelligence, machine learning, computer vision, deep learning, and advanced operating systems.

**MIR SUHAIL ALAM** received the master's degree in computer and information engineering from International Islamic University Malaysia (IIUM). He is currently pursuing the Ph.D. degree in computer science with the King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia. He has professional experience as a Software Developer, a Software Quality Engineer, and a Graduate Research Assistant, and has authored and co-authored several research works in his



**MD SAKIBUL ISLAM** received the B.Sc. degree in computer science and engineering from the Islamic University of Technology, Dhaka, Bangladesh, and the M.Sc. degree from Erasmus Mundus GENIAL. He is currently pursuing the Ph.D. degree in cybersecurity with the King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia.



from the College of Engineering, Department of Computer Engineering and Automatic Control, Tanta University, Egypt. He was an Assistant Professor with the School of Electrical Engineering and Information Science, Kyushu University, from September 2002 to January 2004, before joining KFUPM. He is the principal/co-investigator of many funded research projects and has supervised several Ph.D. students. He has published more than 100 papers in major international journals and conferences. He also edited a book on P2P computing and autonomous agents. His research interests include distributed operating systems, computational intelligence, intelligent agent systems, personalized and semantic web searching, ontology engineering, and cooperative information systems. He serves as a reviewer for several computer science journals and is a member of the organizing committees of international conferences in the areas of artificial intelligence, semantic web, intelligent agents, and distributed systems.

**TAREK HELMY** received the B.S. and M.S. degrees in computer science and engineering from Menoufia University, Egypt, in 1991 and 1995, respectively, and the Ph.D. degree in intelligent systems from Kyushu University, Japan, in 2002. He is currently a Full Professor with the Department of Information and Computer Science, College of Computing and Mathematics, King Fahd University of Petroleum and Minerals (KFUPM). He is also a Full Professor and on leave



Saudi Arabia, with a concentration in networks and cybersecurity.

**MUHAMMAD DIKKO GAMBO** received the B.Sc.TE. degree in computer science and engineering from the Islamic University of Technology, Dhaka, Bangladesh, and the National Innovation Diploma (N.I.D.) diploma in networking and system security from Katsina State Institute of Technology and Management, Katsina, Nigeria. He is currently pursuing the M.Sc. degree in computer engineering with the King Fahd University of Petroleum and Minerals (KFUPM), Dhahran,

...